SegPlantFormer: A Comparative Deep Learning Framework for Plant Semantic Segmentation

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ABSTRACT

With the advent of climate change and a growing population, food security will be a pressing issue in the coming years. To stay ahead, we need tools that help us fully understand the complex processes and functions of specific genes in plant cell biology, and how these genes influence plant growth and morphology. This requires the examination of countless cells or individual plants, especially given the variability in cellular behavior, which becomes critical when studying anomalies or emerging patterns that affect normal cell function and morphology. This task is challenging for individual cell analyses, even with technological advancements. Therefore, we introduce a comparative deep-learning framework for high-content microscopy data analysis, benchmarking Convolutional Neural Networks (CNNs) against Transformer-based architectures to automate the segmentation of the moss *Physcomitrium patens*. Specifically, we evaluate U-Net models with ResNet34 and Transformer (MiT) backbones against a SegFormer baseline. Our results demonstrate that the U-Net (ResNet34) achieves state-of-the-art performance for living plants (CWT) with a mean IoU of 0.756, significantly surpassing human experts (0.491). Conversely, for dead plants, the U-Net (Transformer) proved superior (0.552 mIoU), effectively handling high-noise environments. This study establishes a robust, automated pipeline that distinguishes between living and non-living plants with greater precision than traditional methods. Additionally, we introduce a unique plant cell biology dataset with manually annotated masks.

Introduction

Advances in plant cell biology have been crucial for society due to their significant impact on agriculture and food security, as they provide the opportunity for scientists to develop more productive, disease-resistant, and drought-tolerant crop varieties, thus improving food availability[1]. It also plays a role in environmental sustainability due to the food chain and climate change mitigation through their regulation of carbon dioxide levels[2]. From a fundamental science point of view, plants provide unique opportunities to study essential biological processes such as cell division, cell signaling, and gene regulation[**?**].

The moss *Physcomitrium patens* (P. patens) (**Figure 2**) is an ideal model system for cell biologists because it is easy to handle, yet its molecular and genetic characteristics are well conserved throughout evolution in higher plants[3]. This means that the knowledge gained from studying this plant can be extrapolated to other plants of agricultural interest. Understanding plant biology requires investigating the molecular mechanisms controlling these cellular events and their seamless integration. Phenotypic changes resulting from genetic mutations serve as tools for such investigations. This study takes advantage of the unique attributes of the moss *P. patens* as an ideal model for *in vivo* microscopy due to its compact size, rapid growth, and ease of gene manipulation[4].

However, analyzing high-content microscopy data presents significant challenges. In the big data era, the volume of images makes manual analysis inefficient and prone to human bias[5]. Traditional automated tools, such as ImageJ macros[6], often rely on intensity thresholding, which struggles to differentiate between subtle phenotypic variations, overlapping cells, or distinguishing between living and dead plant tissue in noisy environments[7]. While deep learning has revolutionized biological image analysis[8], most applications in plant biology have focused on disease classification rather than precise cellular segmentation[9,10].

To address this gap, we propose a comprehensive deep learning framework for the semantic segmentation of *P. patens*. Unlike previous approaches that rely on a single architecture, this study conducts a rigorous benchmarking of two leading paradigms in computer vision: Convolutional Neural Networks (CNNs), represented by the U-Net architecture[11], and Transformer-based models, represented by SegFormer[12]. By evaluating these models across varying conditions—specifically focusing on distinguishing healthy (CWT) from dead (Treated) plants—we demonstrate that deep learning significantly outperforms

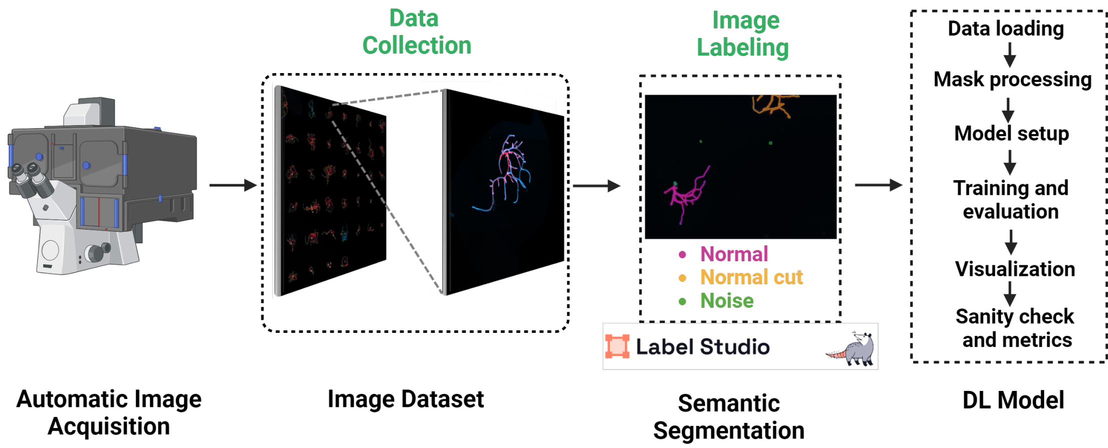

Figure 1. Workflow for Creating a Labeled Image Dataset of P. patens for Semantic Segmentation in Deep Learning Model Training. The workflow begins with automatic image acquisition using a Zeiss microscope, where images of Physcomitrium patens are captured as tiles (10x10). Images are then processed for labeling using the open-source software, Label Studio. Through manual semantic segmentation, each image is annotated with three distinct categories: Normal, Normal cut, and Noise. The annotated labels are generated as masks corresponding to specific image IDs. These labeled datasets, containing both the images and their respective masks, serve as input for training the deep learning model[12]

traditional methods. Furthermore, we provide a unique, expert-annotated dataset designed to advance research in plant phenotyping automation.

Microscopy imaging is frequently utilized in biological and medical studies for cells and tissues. Multiple toolboxes have been implemented for research purposes in model systems like *Caenorhabditis elegans* and zebrafish[13–17]. However, these tools have not been expanded to plant biology in the same way. In the big data era, massive datasets make manual analysis inefficient[5]. Therefore, automation has become particularly appealing, with machine learning methods extensively applied to biological image analysis for tasks like cell type identification, population analysis, complex morphological assays, and perturbation reagents[8]. To study cell morphology and gene-related defects, biologists use high-content microscopy that provides spatiotemporal resolution information crucial for understanding biological system organization[18]. This technique enables detailed analysis of morphological features[?], which is especially important when gene defects result in loss-of-function phenotypes.

There have been attempts to introduce new technologies, such as deep learning tools, in the field of plant biology, primarily for disease detection and recognition[9]. For instance, some models were designed to diagnose tomato plant diseases through leaf image data, while others[19] used the LeNet architecture to identify soybean plant diseases, avoiding the need for segmentation. Similarly,[20] utilized deep learning to detect plant illnesses from images of healthy leaves, while[21] embarked on a two-stage approach for object detection. Other examples include models tailored for mango[22], rice[23], sugar beet[24], and olive plants[25]. While these studies advanced disease detection, they only[10] made significant progress in plant cell segmentation. This highlights a promising area for future research targeting precise plant cell segmentation using deep learning.

Our research uses high-content microscopy data to quantify morphological traits in *P. patens*. To facilitate this, we use ImageJ macros[6] to automate large-scale image processing. The macro identifies and measures objects based on size and intensity[7]. This automation helps identify important phenotypes, especially in RNA interference studies[26]. However, a key step is image segmentation. Given the need to analyze thousands of plants, manual segmentation is inefficient. While traditional algorithms help detect boundaries, errors like over- or under-segmentation can distort cell structures, impacting accuracy[27]. To mitigate this, manual selection is often used but introduces bias. Instead, we implemented a deep learning framework that automatically identifies plants, sets thresholds, and classifies masks, providing faster and more reliable results.

Manual methods and current tools struggle to process large datasets where distinguishing between living plants, dead plants, and noise is essential. To address this, our contributions are:

1. **Unique Data for Plant Semantic Segmentation:** We introduce a ground truth dataset including categories for living and dead plants, as well as noise. This dataset captures important morphological variations in *P. patens*. Each image was carefully labeled using expert knowledge, ensuring reliability for model training.

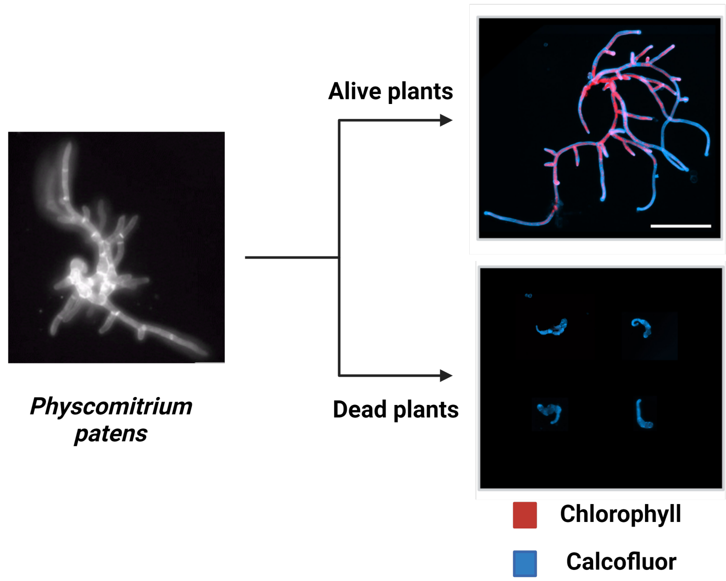

Figure 2. Plant model system *Physcomitrium patens*. On the left panel, the plant is visualized in brightfield imaging, showing its branching morphology. On the right, two panels represent stained images: in the upper panel, normal and healthy plants (alive) are stained with calcofluor (blue) and show red chlorophyll signals, chlorophyll is is an indicator of their vitality. The lower panel shows smaller dead plants, which no chlorophyll signal and were treated with a toxic compound. The dead plants exhibit minimal growth and no branching.

2. **Benchmarking of Deep Learning Architectures:** Unlike previous works that propose a single model, we present a comparative study between Convolutional Neural Networks (U-Net with ResNet34 backbone) and Transformer-based models (SegFormer and U-Net with MiT backbone). This allows us to identify the optimal architecture for distinct biological structures—continuous living plants versus fragmented dead tissue.

3. **Superior Performance and Automation:** We demonstrate that our deep learning framework significantly surpasses traditional ImageJ thresholding and matches or exceeds human expert annotation. Specifically, the U-Net (ResNet) achieved a 53.9% improvement over human annotation in living plants, providing a scalable solution for high-throughput phenotyping (Figure 1).

1 Results and Discussion

1.1 Segmentation Evaluation Metrics

The Intersection over Union (IoU), also known as the Jaccard Index, is a metric widely used in image segmentation benchmark datasets[28–31]. It is particularly well-suited for dense pixel-wise prediction tasks, where the model must classify every pixel in an image. The core concept of IoU is to quantify the overlap between the predicted segmentation and the ground truth relative to their total combined area.

Given a set of classes (e.g., alive, dead, background), each pixel is assigned to a class $c \in \mathscr{C}$. To compute the IoU for a specific class, we follow Eq. 1: the area of the intersection between the predicted segmentation mask $\hat{y}$ and the ground truth mask y is divided by the area of their union.

$$\mathrm{IoU}_c = \frac{|\{y = c\} \cap \{\hat{y} = c\}|}{|\{y = c\} \cup \{\hat{y} = c\}|} \tag{1}$$

In this study, we use the mean IoU (mIoU), averaged across all images in the test set, to evaluate overall model performance and generalization capability.

1.2 Comparison Baselines

To evaluate the effectiveness of our deep learning architectures, we benchmarked them against traditional segmentation workflows used in plant biology:

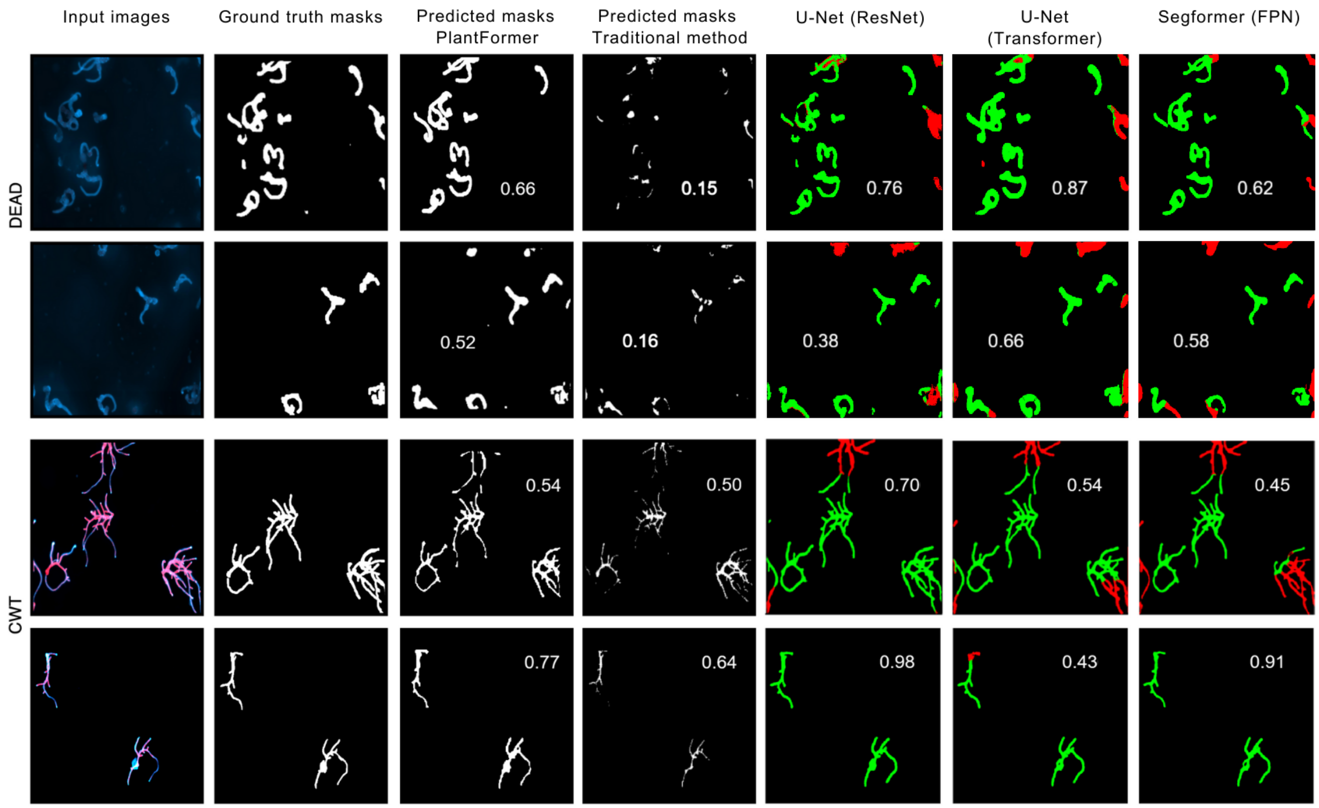

Figure 3. Qualitative comparison of segmentation methods on the Test Set. The figure displays samples from the **DEAD** dataset (top two rows) and **CWT** dataset (bottom two rows). Columns displayed (from left to right): (1) Input Image, (2) Ground Truth, (3) Predictions from the original PlantFormer, (4) Traditional Method, and our three new benchmarked architectures: (5) **U-Net (ResNet)**, (6) **U-Net (Transformer)**, and (7) **SegFormer (FPN)**. The Intersection over Union (IoU) score is overlaid on each prediction. **Key Observation:** Note how the **U-Net (ResNet)** achieves superior boundary definition in the continuous branching structures of CWT (e.g., Row 4, IoU 0.988), whereas the **U-Net (Transformer)** excels in capturing the fragmented morphology of DEAD plants (e.g., Row 1, IoU 0.870), significantly outperforming the Traditional Method and the original baseline. Each input image is labeled with a corresponding ID, matching the results reported in Table 2.

- **Baseline (Automated ImageJ):** This represents the traditional automated method using a custom ImageJ macro. The workflow includes a Gaussian Blur filter to smooth the image and reduce noise, followed by contrast enhancement to optimize the intensity range for standard thresholding.
- **Human Expert (Semi-automated):** These masks were generated using the same ImageJ macro but were subsequently refined manually by a domain expert to correct segmentation errors, serving as a high-quality "human-level" benchmark.

We compared these baselines against the masks predicted by our proposed deep learning models (U-Net variants and SegFormer), as visualized in **Figure 3**.

1.3 Model Results

To identify the most effective architecture for plant phenotyping, we trained and evaluated three distinct deep learning models on both the CWT and Dead datasets: (1) **SegFormer (FPN)** with a MiT-B2 backbone, (2) **U-Net** with a ResNet34 backbone, and (3) **U-Net** with a Transformer (MiT-B2) backbone. These were benchmarked against our original SegFormer (MiT-B0) baseline.

Training Configuration. To ensure a fair comparison, all new models were trained using a unified setup: 70 epochs, a batch size of 8, and an AdamW optimizer with a learning rate of $1e-4$ and a Cosine Annealing scheduler. Unlike previous iterations, we employed a **Joint Loss** function (50% Dice Loss + 50% Focal Loss) to better handle class imbalance between plant pixels and the background. Data augmentation techniques, including elastic deformations and geometric transformations, were applied to improve generalization.

Quantitative Analysis. Table 1 summarizes the performance of all methods. The results demonstrate a clear architectural trade-off based on biological morphology:

- **CWT (Living Plants):** The **U-Net (ResNet34)** achieved state-of-the-art performance with a mean IoU of **0.756**, surpassing the Human Expert (0.491) by a margin of 53.9%. This indicates that Convolutional Neural Networks are highly effective at delineating the continuous, well-defined boundaries of healthy plants.
- **Dead (Treated Plants):** In this challenging scenario characterized by fragmentation and noise, the **U-Net (Transformer)** emerged as the superior model with an mIoU of **0.682**. Its self-attention mechanism likely enables it to aggregate context from disjointed cellular debris better than CNN-based approaches.

Table 1. Comparative Benchmarking Results. We compare the Mean Intersection over Union (mIoU) across traditional methods and deep learning architectures. The U-Net (ResNet34) excels in the healthy (CWT) dataset, while the U-Net (Transformer) dominates in the noisy (Dead) dataset. Both outperform the Human Expert benchmark.

Method	Backbone	CWT (Alive) mIoU	Dead (Treated) mIoU
Traditional Methods			
Baseline (ImageJ)	N/A	0.421	0.256
Human Expert	N/A	0.491	0.404
Deep Learning Models			
SegFormer (Original)	MiT-B0	0.535 ± 0.005	0.428 ± 0.029
SegFormer (FPN)	MiT-B2	0.637 ± 0.011	0.637 ± 0.015
U-Net (ResNet)	**ResNet34**	$\mathbf{0.756} \pm 0.016$	0.605 ± 0.007
U-Net (Transformer)	**MiT-B2**	0.673 ± 0.009	$\mathbf{0.682} \pm 0.028$

Per-Image Analysis. As detailed in the supplementary material (Table 2), these improvements are consistent across individual test images. For instance, in the CWT dataset, the U-Net (ResNet) improved segmentation accuracy in almost all samples compared to traditional methods, successfully capturing fine branching structures that were previously missed by thresholding techniques.

1.4 Impact of Automated Phenotyping

When studying the intricate processes within cells, researchers face the challenge of examining vast numbers of individuals. Cells do not exhibit uniform behavior, especially under gene depletion or drug treatments. Ideally, every cell would be analyzed individually, but the scale makes manual analysis unfeasible.

With high-content microscopy and our proposed deep learning framework, we can now assess phenotypes automatically. This approach boosts reliability and enables comprehensive analysis, avoiding the bias of selecting only "representative" images. Our findings address three critical challenges:

1. **Overlapping:** Cell overlap complicates parameter measurement. Deep learning models, particularly U-Net architectures, showed improved separation of adjacent structures compared to intensity-based thresholding.
2. **Classification Accuracy:** Manual selection ("shaving") is labor-intensive. Our automated models achieve segmentation quality that matches or exceeds human annotation, reducing the need for manual intervention.
3. **Differentiation of Cell States:** Distinguishing between living and dead cells is difficult for traditional macros. Our results show that specialized architectures (CNNs for living, Transformers for dead) can reliably automate this distinction.

Implementing these models revolutionizes plant cell growth studies by increasing throughput and consistency.

1.5 Discussion

Implementation Advantages. The proposed framework offers significant benefits over traditional ImageJ macros. By validating multiple architectures, we provide a tailored solution: U-Net (ResNet) for standard growth assays (CWT) and U-Net (Transformer) for complex mortality assays (Dead). This flexibility allows researchers to process extensive datasets with higher accuracy (up to 75.6% mIoU) than manual experts, ensuring reproducible results critical for quantitative phenotyping.
Limitations. While effective, the models are fine-tuned for *P. patens*. Applying them to species with vastly different morphologies may require re-training. Additionally, extremely noisy images (like ID 10 in Table 2) remain a challenge

Table 2. Detailed Performance by Test Image. We report the Intersection over Union (IoU) for individual images in the test set. The **U-Net (ResNet34)** consistently dominates in the CWT dataset, while the **SegFormer** and **U-Net (Transformer)** show competitive performance in the difficult Dead dataset scenarios.

ID	Dataset	Filename	Baseline	HumanExpert	SegPlantFormer	U-Net (ResNet)	U-Net (Transf)	SegFormer (FPN)
1	CWT	cwt3_6.jpg	0.619	0.675	0.682	**0.885**	0.870	0.399
2	CWT	cwt2_65.jpg	0.229	0.403	0.469	0.318	**0.770**	0.660
3	CWT	cwt1_43.jpg	0.571	0.637	0.755	0.874	**0.966**	0.868
4	CWT	cwt4_3.jpg	0.003	0.003	0.004	**0.608**	0.608	0.597
5	CWT	cwt2_23.jpg	0.374	0.420	0.577	**0.873**	0.731	0.850
6	CWT	cwt2_15.jpg	0.424	0.609	0.293	**0.714**	0.203	0.034
7	CWT	cwt3_75.jpg	0.498	0.546	0.542	**0.706**	0.548	0.457
8	CWT	cwt1_75.jpg	0.635	0.762	0.774	**0.988**	0.436	0.919
9	CWT	cwt2_76.jpg	0.254	0.279	0.385	0.788	0.688	**0.788**
10	CWT	cwt1_73.jpg	0.601	0.572	0.701	0.805	**0.907**	0.794
1	Dead	dead1_29.jpg	0.146	0.669	0.641	0.763	**0.870**	0.618
2	Dead	dead1_3.jpg	0.000	0.008	0.003	**0.823**	0.484	0.644
3	Dead	dead2_42.jpg	0.520	0.520	0.530	0.439	**0.566**	0.562
4	Dead	dead1_40.jpg	0.158	0.370	0.493	0.382	**0.662**	0.582
5	Dead	dead2_26.jpg	0.455	0.455	0.630	0.617	**0.826**	0.780

for all models, yielding very low IoU scores. This suggests that some "noise" labeled in the dataset might be practically indistinguishable from background, or that further data augmentation is needed for these edge cases.

Impact in Cell Biology. By offering precise, automated segmentation, this framework allows for population-wide analysis rather than limited sampling. This is crucial for identifying subtle morphological variations and emergent behaviors that traditional, manual methods often overlook due to sample size limitations.

1.6 Future Work

Future research will prioritize transitioning from semantic segmentation to **instance or panoptic segmentation**. While our current models successfully classify pixels as "living" or "dead," distinguishing individual plant instances within dense clusters remains a challenge due to overlapping. Adopting architectures capable of instance discrimination would allow for precise plant counting and automated morphological tracking of individual organisms.

Another critical direction is the integration of **multi-modal data**. Combining fluorescence channels with phase-contrast or brightfield imaging could provide complementary feature sets, potentially resolving ambiguities in highly noisy environments where the chlorophyll signal is faint or absent.

Furthermore, we plan to conduct rigorous **latency and efficiency studies**. Given the architectural differences between CNNs (typically faster inference) and Transformers (higher computational cost), assessing their trade-offs is essential for real-time processing. This would pave the way for embedding these models directly into microscope software for "on-the-fly" phenotyping during image acquisition.

Finally, expanding the training data to include **loss-of-function phenotypes** and stress conditions will be vital. By training on subtle morphological variations—such as changes in branching angles or cell length—the models could uncover emergent biological behaviors that are currently undetectable by manual analysis.

2 Conclusion

In this paper, we presented a comparative deep-learning framework for the semantic segmentation of the model system *P. patens* using high-content microscopy data. By benchmarking Convolutional Neural Networks (U-Net ResNet) against Transformer-based architectures (SegFormer and U-Net MiT), we identified that optimal architecture selection depends heavily on the biological morphology of the sample. Our results show that while CNNs excel at delineating the continuous structures of living plants, Transformers are superior for identifying fragmented dead tissue in noisy environments. Our approach significantly outperforms traditional segmentation methods and human expert annotation, offering a highly efficient, scalable, and automated solution. This framework serves as a robust baseline for future automated image analysis in plant biology research.

3 Method

To explore gene function in *P. patens*, we designed an experimental pipeline that integrates microscopy and automated image processing. We performed protoplast transformation followed by high-throughput imaging of the resulting plants. This automated workflow enables us to process large datasets, ensuring accuracy in identifying phenotypic variations while reducing the errors associated with manual or semi-automated methods.

3.1 Experimental Pipeline

Moss Culture and Transformation: The experiment begins with protoplast transformation, where the moss cell walls are enzymatically digested to produce protoplasts—cells without their rigid walls. This step is crucial for efficient plasmid uptake (genetic treatment). We used an RNA interference vector system to silence specific genes (depletion of function). The transformed protoplasts regenerate for four days, followed by another four days of selection treatment. By the end of this period, hundreds of individual plants are produced, each potentially showing phenotypes linked to the silenced gene, providing a valuable dataset for analysis.

Staining Protocol: To clearly visualize the plant cells, we applied Calcofluor-White, a fluorescent dye that binds to cellulose in the plant cell walls, for 2 minutes before taking images. This staining highlights the cell boundaries, which is essential for accurate segmentation during image analysis.

High-Resolution Image Acquisition and Stitching: Seven days post-transformation, we captured multi-dimensional images of the moss cultures using an epifluorescence microscope. We captured 225 individual image tiles across two channels—red for chlorophyll and blue for cell walls stained with Calcofluor-White. These high-resolution images capture the diversity within the plant population, which is crucial for studying morphological changes resulting from living, dead, gene silencing, and noise. The captured image tiles were stitched together to create composite images that cover the entire culture.

Image Processing: The stitched images were then processed in ImageJ, where a custom macro automated the segmentation and thresholding steps. This process extracted key morphological features like area, solidity, and intensity, which are important for identifying phenotypic variations.

Challenges and Limitations: The main challenges in this experiment were ensuring the accuracy of segmentation and thresholding, particularly when differentiating between living and dead plants or filtering out background noise. The results can vary between experiments. In addition, despite the high quality of the images, overlapping plants and artifacts made the analysis more difficult, highlighting the need for advanced deep learning tools to improve segmentation accuracy.

3.2 Deep Learning Architectures

We formalized the problem as a semantic segmentation task, where the goal is to classify every pixel p_{ij} into a semantic class $c \in \{\text{background, living, dead}\}$. Unlike previous approaches relying on a single model, we implemented a comparative framework using the `segmentation-models-pytorch` library to benchmark Convolutional Neural Networks against Transformer-based architectures.

3.2.1 Convolutional Approach: U-Net (ResNet34)

As a baseline for CNN performance, we employed the U-Net architecture[11]. We replaced the standard encoder with a ResNet34 backbone pre-trained on ImageNet. This architecture relies on convolutional operations that excel at capturing local texture and defining continuous boundaries, which is hypothesized to be beneficial for the well-defined structures of living plants (CWT).

3.2.2 Transformer-based Approach: SegFormer and U-Net (MiT)

To capture global context, we utilized the Mix Transformer (MiT) encoders, originally proposed in SegFormer[12]. These encoders split the image into patches and process them via hierarchical self-attention mechanisms. We tested two variants:

- **SegFormer (FPN):** Uses a lightweight All-MLP decoder with a Feature Pyramid Network (FPN) to aggregate multi-scale features.

- **U-Net (Transformer):** Uses the MiT-B2 encoder but feeds the features into a U-Net decoder with skip connections, allowing for finer reconstruction of spatial details[32].

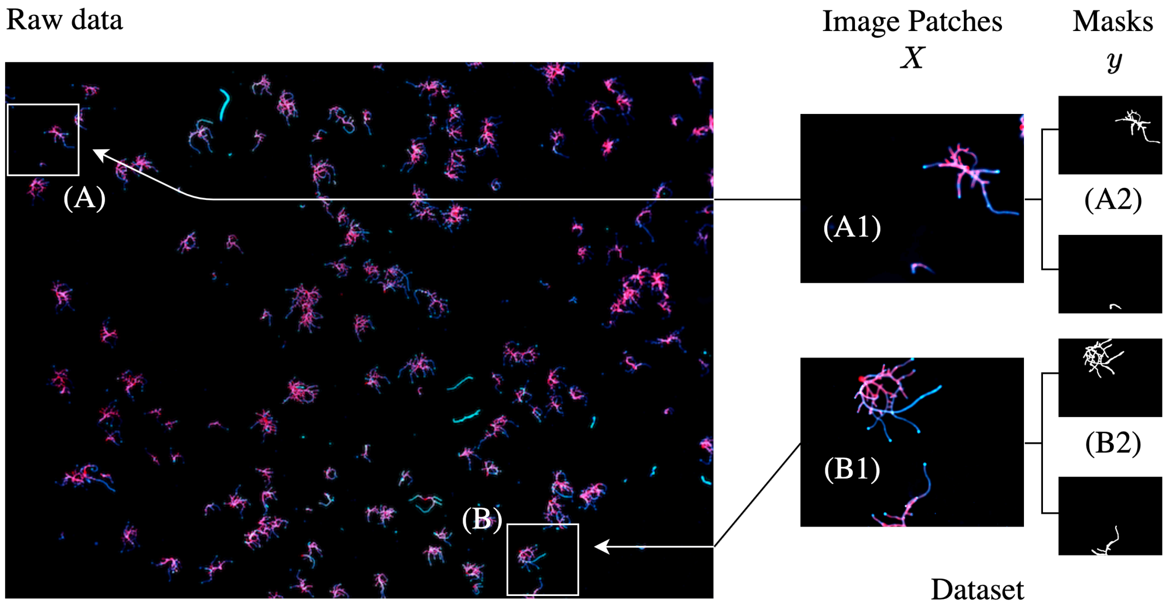

Figure 4. Dataset collection process. The results of the laboratory experiments are represented by a high-resolution stacked image captured using microscopic imaging (Raw Data). The white squares labeled **(A)** and **(B)** indicate the regions from which samples were extracted. Smaller image patches extracted from these samples, labeled **(A1)** and **(B1)**, serve as the input for the model (X). Masks with semantic information, such as **(A2)** and **(B2)**, are manually created to annotate the plant shapes with corresponding alive/dead cell labels, forming the target y. Note that each image may contain multiple masks depending on the number of entities present within the image patches.

3.2.3 Training Strategy

To improve model robustness, we implemented a **Joint Loss** function combining Dice Loss and Focal Loss (weighted 0.5 each) to mitigate class imbalance. Furthermore, we applied an aggressive data augmentation pipeline using the Albumentations library, incorporating elastic transformations and grid distortions to simulate natural morphological variability. Models were trained for 70 epochs using the AdamW optimizer.

3.3 Dataset Collection and Labeling

Dataset Collection. The image dataset originates from plant cell growth assays specifically designed to study the phenotype of the moss *P. patens*. The dataset involves the growth patterns of thousands of cells after an 8-day growing period. We curated images from three independent experiments: the **"Control WildType (CWT)"** dataset (living cells) and the **"Dead"** dataset (cells artificially induced to die). Any background artifacts were categorized under the "noise" label.
Data Labeling Pipeline. To create the ground truth ($\mathscr{Y}$), we followed a systematic patching strategy (Figure 4). Large microscopy tiles were divided into smaller, manageable patches (e.g., 2519×1895 pixels) using the Python package "split-image"[33]. Using Label Studio[34], we manually annotated semantic masks for each patch. This process resulted in a high-quality dataset of paired images ($\mathscr{X}$) and masks ($\mathscr{Y}$), which were subsequently split into training (85%) and validation (15%) sets.

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